B. Tech. Fifth Semester (Information Technology) / (SOE_23 IT-101) Examination

Course Code : 23 IT 1501 **Course Name : Theory of Computation**

Time : 3 Hours] [Max. Marks : 50

Instructions to Candidates :—

(1) Do not write anything on question paper except your exam seat number.

(2) Write the accurate question number in left margin of answer book along with answers.

1. Solve any **Two** :

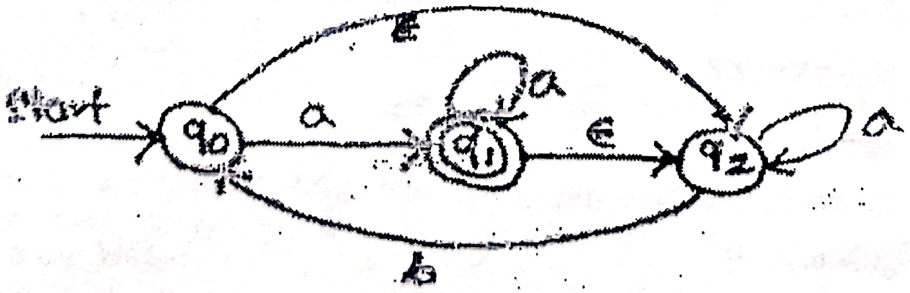
(a) Convert the following NFA to its equivalent DFA with minimization q**0** be the initial state and q**2** be the final state :

Q \ Σ	a	b
→ q_0	$\{q_0 q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
* q_2	—	$\{q_0 q_1\}$

4.5(CO1)

(b) Give the difference between a NFA and DFA with suitable example. 4.5(CO1)

(c) Convert the following NFA with 'ε' - Transition to its equivalent NFA without 'ε' - transition.



4.5(CO1)

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2. Solve any **Two** :

(a) Write regular expression for the following finite automata using Arden's theorem.



4(CO2)

(b) Construct Non-Deterministic Finite Automata with ε-moves for Regular Expression :

a . (a + b)* . b . b

4(CO2)

(c) Represent the following set by regular expression :

(i) The set of all strings over {0 , 1} with at least two 0's.

(ii) The set of all strings over {a , b} beginning and ending with b.

4(CO2)

3. Solve any **Two** :

(a) Eliminate ε production from the grammar G , G is consisting of following grammar :

(i) S→ aSa | bSb | ε

(ii) S→ a | Xb | aYa
X→ y | ε

4(CO2)

(b) Simplify the following Grammar by eliminating the Useless Grammar symbol from the grammar :

A→ xyz | Xyzz

X→ Xz | xYx

Y→ yYy | XZ

Z→ Zy | z

4(CO2)

(c) Convert the following CFG into CNF form :

S→ bA | aB

A→ bAA | aS | a

B→ aBB | bS | b

4(CO2)

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4. Solve any **Two** :

(a) Design a Push Down Automata for $L = \{0^n 1^n \mid n >= 1\}$ is accepted by empty stack. 4.5(CO4)

(b) Construct Push Down Automata for $L = \{a^n b^m c^n \mid n >= 1, m >= 1\}$. 4.5(CO4)

(c) Construct a PDA for the following CFG and test whether "abbabb" is accepted or not :
$G = (\{A\}, \{a, b\}, R, S)$
$R = \{S \rightarrow aAA, A \rightarrow aS \mid bS \mid a\}$ 4.5(CO4)

5. Solve any **Two** :

(a) Design a Turing Machine to find 2's Compliment of a number. 4(CO4)

(b) Prove that : "If L1 and L2 are two recursive languages then the Union of L1 and L2 is a recursive Language". 4(CO4)

(c) Design a Turing Machine for which it recognizes the language :
$L = \{0^n 1^n \mid \text{where } n >= 1\}$ 4(CO4)

6. Solve any **Two** :

(a) Decide whether the following (A, B) pairs have a Post Correspondence Solution or not ? If yes give a solution, if no why ?

	List A	List B
i	wi	xi
1	0	000
2	01000	01
3	01	1

4(CO3)

(b) Compute $A(1,1), A(2,1), A(1,2), A(2,2)$ using Ackermann's function. 4(CO3)

(c) Trace how a **Universal Turing Machine** simulates a Turing Machine M on input w and explain why this simulation implies the undecidability of certain languages. 4(CO3)

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Hingna Road, Wanadongri, Nagpur.

Department of Information Technology

ODD Term—2025-26
Semester – V
Subject Code : 23IT1501

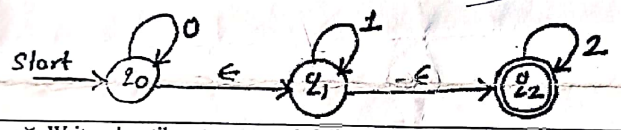
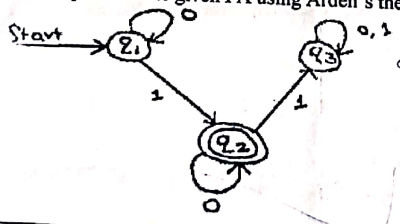
Mid Semester Exam I
Date: - 12/09/2025
Subject Theory of Computation

Time :1½ Hours

Max. Marks: 30

Instructions:

1. All questions are compulsory
2. Give appropriate figure where ever necessary
3. Marks will be allotted as zero if wrong question number is mentioned.

Q.1.(Any Two)	a) Design a DFA ending with 101, ∑={0,1}	[05]	CO2 (Level 4)
	b) Construct DFA equivalent to the following M= ({q0, q1, q2, q3}, {a,b}, δ, q0,{ q3}) where δ is given as 5 δ | 0 | 1 q_0 | $\{q_0, q_1\}$ | q_0 q_1 | q_2 | q_1 q_2 | q_3 | q_3 $*q_3$ | - | q_2	[05]	CO2 (Level 2)
	c) Convert the following NFA with ε to NFA without ε moves 	[05]	CO2 (Level 2)
Q.2.(Any Two)	a) Write algortihm to convert left linear grammar to right linear grammar. Apply the algortihm on given grammaer to convert it to right linear grammar. S→Aa/ a/ b A→Sb/ Ca / a C→c	[05]	CO2 (Level 2)
	b) Find regular expression for given FA using Arden's theorem. 	[05]	CO2 (Level 2)
	c) List and explain types of grammars in the Chomsky Hierarchy?	[05]	CO2 (Level 1)
Q.3. (Any Two)	a) Consider the grammar given as, G=({S, A}, {a, b}, P, S) where 'P' consisting of, S→AaS/a A→SbA/SS/ba Derive "aabbaa" using left most and rightmost derivation. Also draw left most and rightmost derivation tree.	[05]	CO2 (Level 2)
	b) What is ambiguous grammar? Check whether the given grammar is ambiguous or not- S → SS S → a S → b Consider w = abba	[05]	CO2 (Level 3)
	c) How to remove useless symbol from the grammar? Test whether the grammar contains useless symbol, if yes eliminate it. S→aS/ A/ C A→a B→aa C→aCa	[05]	CO2 (Level 3)

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Department of Information Technology

ODD Term—2025-26
Semester – V
Subject Code : 23IT1501

Mid Semester Exam II
Date: - 13/11/2025
Subject Theory of Computation

Time :1½ Hours

Max. Marks: 30

Instructions:

1. All questions are compulsory
2. Give appropriate figure where ever necessary
3. Marks will be allotted as zero if wrong question number is mentioned.

Q.1.(Any Two)	a) Design Push Down Automata for the language $L = \{ a^n b^{2n} \mid n>=1$ by empty stack}	**[05]**	CO4 (Level 6)
	b Construct Push Down Automata for the following context free grammar S→ 0BB B→ 0S | 1S | 0 Test whether "010000" is in N (A)	**[05]**	CO4 (Level 6)
	c) Design Push Down Automata for the language L L = {x | x consists of equal number of a's and b's}.	**[05]**	CO4 (Level 6)
Q.2.(Any Two)	a) Construct Turing Machine that can accept all strings over {a, b} containing odd Number of a's.	**[05]**	CO4 (Level 6)
	b) Design Turing Machine to find 2's Complement of given binary number.	**[05]**	CO4 (Level 6)
	c) Show that the union of two recursive y enumerable languages is recursively enumerable	**[05]**	CO4 (Level 3)
Q.3. (Any Two)	a) Write Ackerman function. Apply this function to compute A (1, 3) and A (2, 1).	**[05]**	CO3 (Level 3)
	b) When a language is said to be recursive? Is it true that every regular set is not recursive	**[05]**	CO3 (Level 3)
	c) State Post Correspondence problem. Find whether the lists A=(11,100,111) and B= (111, 001, 11) have a Post Correspondence Solution ?Write its applications.	**[05]**	CO3 (Level 3)

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